

ID:

Name:

**Brac University**

Semester: Fall 2025

Course Code: CSE251

Electronic Circuits And Devices

Section: 07

Faculty: PDS

Assessment: Quiz 4

Duration: 30 minutes

Date: January 07, 2025

Full Marks: 20

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ Proper units must be included for all calculated values. Marks will be deducted for missing or incorrect units.
- ✓ Symbols have their usual meanings.

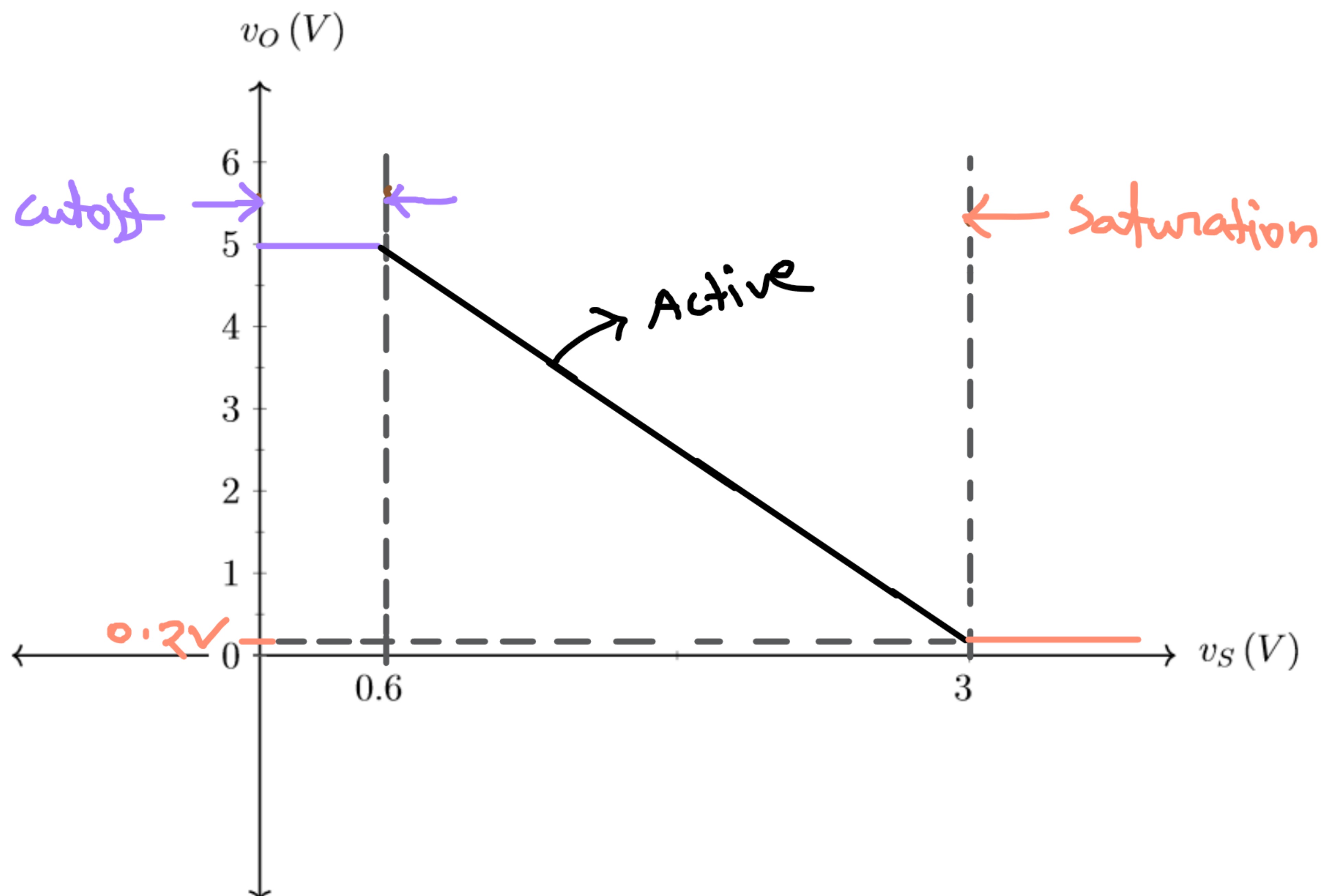
■ Question 1 of 2

[CO1] [5 marks]

For an inverter circuit utilizing an NPN BJT, the output voltage v_O is expressed as a function of the input voltage v_I by the following piecewise relationship:

$$v_O = \begin{cases} 5 \text{ V}, & v_I \leq 0.6 \text{ V}, & \text{Cutoff} \\ -2v_I + 6.2, & 0.6 \text{ V} < v_I \leq 3 \text{ V}, & \text{Active} \\ 0.2 \text{ V}, & v_I > 3 \text{ V}, & \text{Saturation} \end{cases}$$

- (a) [2 marks] Identify and write the corresponding mode of operation (Cut-off, Active, or Saturation) in the boxes provided above.
- (b) [3 marks] Plot the Voltage Transfer Characteristics (VTC) for this inverter on the grid provided below. Specific the operating regions on your graph.



■ Question 2 of 2

[CO2] [15 marks]

For the npn transistor shown below, $\beta = 100$, $V_{BE, active} = 0.7\text{ V}$, $V_{BE, sat} = 0.8\text{ V}$, and $V_{CE, sat} = 0.2\text{ V}$. Determine whether the base-collector junction is forward-biased or reverse-biased.

If we assume the transistor is in Active region,

$$V_B = 0\text{ V}; \quad V_{BE} = 0.7\text{ V}$$

$$\Rightarrow V_B - V_E = 0.7\text{ V}$$

$$\Rightarrow V_E = -0.7\text{ V}$$

$$I_E = \frac{V_E - (-10)}{4.65} = 2\text{ mA}$$

$$I_B = \frac{I_E}{\beta + 1} = 0.0198\text{ mA}$$

$$I_C = \beta I_B = I_B + I_C = 1.98\text{ mA}$$

$$\Rightarrow \frac{5 - V_C}{3} = 1.98\text{ mA}$$

$$\Rightarrow V_C = -0.94\text{ V}$$

$$\text{Condition check: } V_{CE} = V_C - V_E = -0.94 - (-0.7) = -0.24 < V_{CE(sat)}$$

So, it cannot be in active mode.

Now, assuming the transistor is operating in saturation,

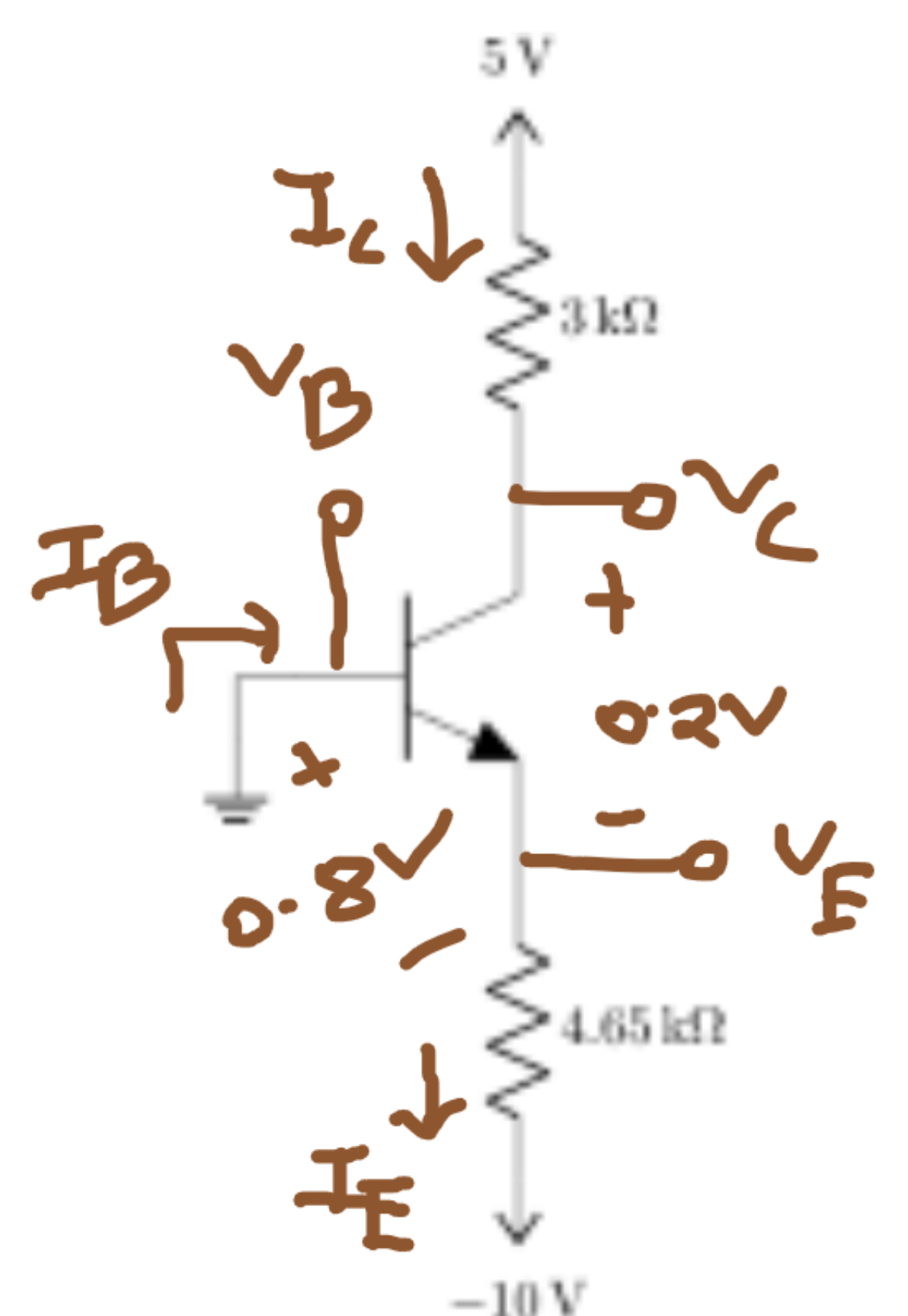
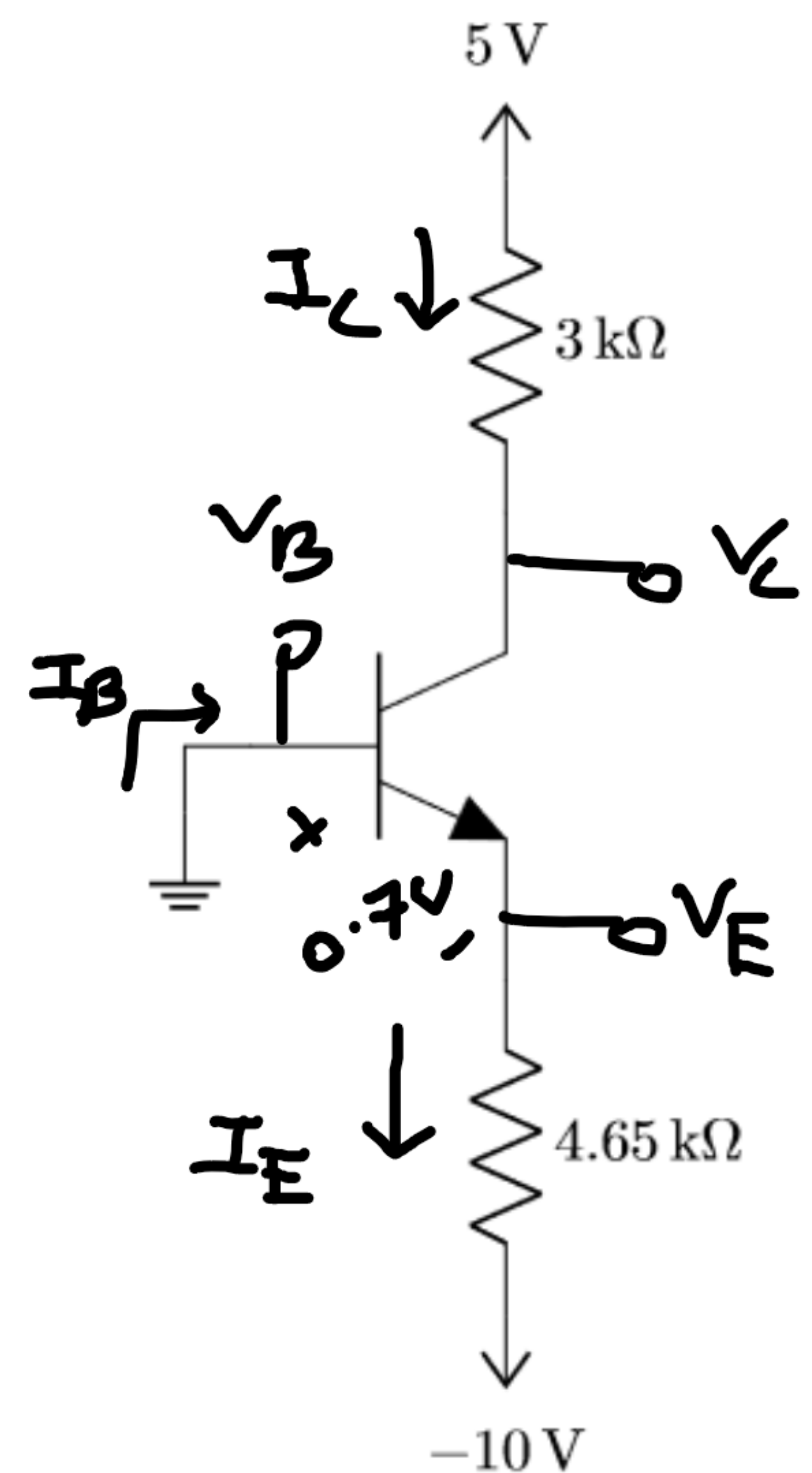
$$V_{BE} = 0.8\text{ V} \neq V_B = 0\text{ V}. \text{ So, } V_E = -0.8\text{ V}$$

$$I_E = \frac{-0.8 - (-10)}{4.65} = 1.978\text{ mA}$$

$$\text{Again, } V_{CE(sat)} = 0.2\text{ V} = V_C - V_E$$

$$\Rightarrow V_C = -0.6\text{ V} [V_E = -0.8\text{ V}]$$

$$\frac{5 - V_C}{3} = I_C = 1.867\text{ mA}$$



$$I_B = I_E - I_C = 0.111 \text{ mA}$$

$$\text{Condition check: } \frac{I_C}{I_B} \approx 16 < \beta$$

so, the transistor is in saturation region and so B-C junction should be forward biased.

$$V_{BC} = 0 - (-0.6) = 0.6 \text{ V } \checkmark$$